SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SIM — Security Incident Mapping Platform

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1. Introduction

1.1 Purpose

This document specifies the software requirements for SIM (Security Incident Mapping), a web-based platform that enables students, staff, and landlords within the AAUA campus community to quickly and safely report security incidents such as theft, assault, or suspicious activity, in order to support faster response and reduction of campus community crime.

1.2 Scope

SIM allows registered and anonymous users to submit incident reports with a location, category, description, and optional evidence. Reports are geotagged/pinned and displayed on a live incident map and list for campus security personnel and the community to view. The initial release (this submission) is a functional minimal viable product: incident submission, incident listing/map view, and basic categorisation. Authentication, notifications, and analytics are noted as future scope.

1.3 Intended Audience

Students and villa mates residing on/off campus

University staff and campus security officers

Landlords housing students off-campus

Course instructor/assessor (Dr. O. O. Ajayi) reviewing the deliverable

1.4 Definitions & Abbreviations

Term	Definition
SIM	Security Incident Mapping — the system being specified
Incident	A reported security event (e.g. theft, assault, stalking, fire, other)
SRS	Software Requirements Specification
MVP	Minimum Viable Product

2. Overall Description

2.1 Product Perspective

SIM is a new, standalone web application. It is not a modification of an existing system. It consists of a client-facing web interface, a backend API/data store for incident records, and a map-based visualisation layer.

2.2 Product Functions (Summary)

Submit a security incident report (category, location, description, date/time, optional anonymity)

View all reported incidents on a map and as a filterable list

Filter/search incidents by category, date, or location

Basic admin view for campus security to review reports

2.3 User Classes and Characteristics

User Class	Description	Technical Skill
Reporter (Student/Staff/Landlord)	Submits incident reports	Basic
Viewer	Browses the incident map/list for awareness	Basic
Campus Security/Admin	Reviews and manages reported incidents	Basic–Intermediate

2.4 Operating Environment

Client: Any modern web browser (Chrome, Edge, Firefox) on desktop or mobile

Server: Static hosting / lightweight backend (Node.js or equivalent), hosted via GitHub

No native mobile app required for this MVP; interface is responsive

2.5 Design and Implementation Constraints

Submission timeline constraint: mini web interface must avoid unnecessary complexity

Must be deployable/demonstrable from a GitHub repository

Data sensitivity: incident data must be handled with care given its sensitive nature

2.6 Assumptions and Dependencies

Users have internet access and a web-enabled device

Location data is entered manually (campus location name) for this MVP rather than live GPS

3. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow a user to submit an incident report containing: category, description, location, and date/time.	High
FR-02	The system shall allow a reporter to choose to submit anonymously or with contact details.	High

FR-03	The system shall display all submitted incidents in a list view, most recent first.	High
FR-04	The system shall display submitted incidents as markers/pins on a campus map view.	Medium
FR-05	The system shall allow users to filter incidents by category and/or date range.	Medium
FR-06	The system shall validate that required fields (category, location, description) are completed before submission.	High
FR-07	The system shall timestamp every report automatically at the point of submission.	High
FR-08	The system shall provide a confirmation message to the user after successful submission.	Medium

4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	The interface shall be simple enough for a first-time user to submit a report within 2 minutes without instructions.
NFR-02	Performance	The incident list/map shall load within 3 seconds under normal campus network conditions.
NFR-03	Availability	The system shall be accessible 24/7 to allow incident reporting at any time.
NFR-04	Security & Privacy	Anonymous reports shall not store or expose any personally identifying information.
NFR-05	Reliability	Submitted reports shall not be lost between submission and storage under normal operation.
NFR-06	Portability	The system shall be accessible from both desktop and mobile browsers (responsive design).
NFR-07	Maintainability	Code shall be version-controlled via GitHub with clear commit history for future extension.
NFR-08	Scalability	The data model shall support future addition of authentication and notification features without redesign.

5. System Architecture Overview

SIM follows a simple three-tier architecture:

Presentation Layer — HTML/CSS/JavaScript front-end providing the report form, incident list, and map view.

Application Layer — handles form validation, request routing, and business logic for creating/retrieving incident records.

Data Layer — stores incident records (category, description, location, timestamp, anonymity flag) in a lightweight data store (e.g. JSON store or database) for this MVP.

Component interaction: The Reporter interacts with the Presentation Layer to submit a report → the Application Layer validates and timestamps the data → the Data Layer persists the record → the Presentation Layer queries the Application Layer to render the updated list/map for all Viewers and Admins.

6. Use Case Summary

Full diagrams are provided separately (use_case_diagram.png / .drawio). Key use cases:

UC-1: Submit Incident Report (Actor: Reporter)

UC-2: View Incident Map/List (Actor: Viewer, Reporter, Admin)

UC-3: Filter Incidents (Actor: Viewer, Admin)

UC-4: Review/Manage Reports (Actor: Campus Security/Admin)

7. Future Scope (Out of Current MVP)

User authentication and role-based access control

Real-time push notifications to campus security

Live GPS-based geotagging

Analytics dashboard for crime-pattern trends (linking back to the Ishikawa root-cause analysis)

8. Appendix

Related artefacts submitted alongside this SRS in the GitHub repository:

Ishikawa (Fishbone) root-cause analysis diagram

Use-case diagram(s)

Stakeholder questionnaire (Google Form) and summary of intended respondents
Mini web interface source code